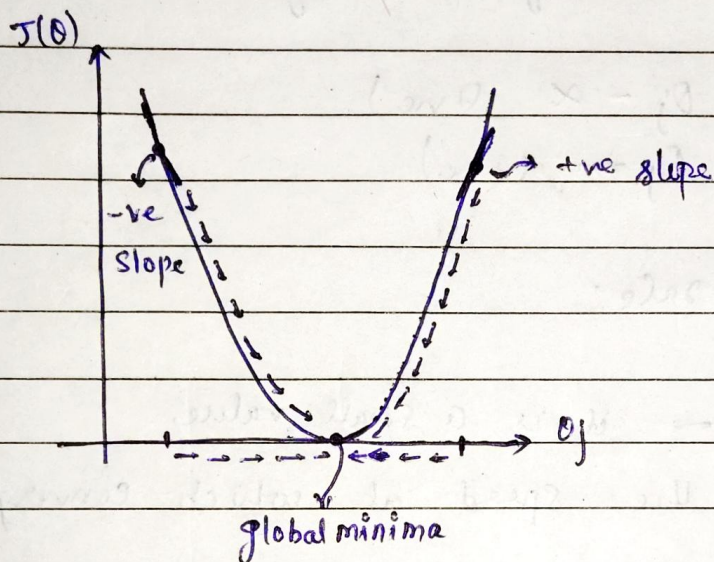


* Convergence Algorithm

{ optimize the changes of θ_i values }

Repeat until convergence

$$\left\{ \begin{aligned} \theta_j &= \theta_j - \alpha \left\{ \frac{\partial J(\theta_j)}{\partial \theta_j} \right\} \rightarrow -ve \\ \end{aligned} \right\}$$



Note:- This equation change the θ_i value much more efficiently.

Make or draw a tangent from the point and if the right side is downward then it is -ve slope

If you want to find the derivative:

$$\left\{ \frac{\partial J(\theta_j)}{\partial \theta_j} \right\} \rightarrow \text{The value will be -ve.}$$

If -ve

$$\left\{ \begin{aligned} \theta_j &= \theta_j - \alpha (-ve) \\ \theta_j &= \theta_j + (+ve) \end{aligned} \right\}$$

↪ value

Means I have to increase the θ_j value. So, if we repeat this process again and again and we found -ve slope then we have to increase the θ_j value in order to reach the global minima.

$$\theta_j = \theta_j - \alpha \left[\frac{\partial}{\partial \theta_j} J(\theta_j) \right] \rightarrow \underline{+ve}$$

+ve slope when right side of the slope going upward then we have to decrease the θ_j value and repeat the process again and again to reach the global minima.

$$\begin{aligned} \theta_j &= \theta_j - \alpha (+ve) \\ &= \theta_j - (-ve) \end{aligned}$$

α = learning rate

let

$\alpha = 0.001 \rightarrow$ it is a small value.

$\hookrightarrow$ controls the speed at which convergence happen.

If value is very very small then it takes more time to converge and if it is big then continuously jumps here and there and never converge to

$\alpha = 0.001 \rightarrow$ Best Practice